

TABLE I: Detailed case-level results for neural architecture search, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). Blue denote the best optimizer for each SE case.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
c10mop10_0	2.0	2.0	1.0	2.0	0.210 $\pm$ 0.113	0.430 $\pm$ 0.289	0.287 $\pm$ 0.275	0.315 $\pm$ 0.324
c10mop11_0	1.0	3.0	2.0	1.0	0.610 $\pm$ 0.000	0.610 $\pm$ 0.158	0.527 $\pm$ 0.345	0.361 $\pm$ 0.343
c10mop12_0	2.0	2.0	1.0	2.0	0.519 $\pm$ 0.000	0.481 $\pm$ 0.048	0.404 $\pm$ 0.320	0.500 $\pm$ 0.047
c10mop13_0	4.0	1.0	3.0	2.0	0.760 $\pm$ 0.079	0.613 $\pm$ 0.260	0.748 $\pm$ 0.120	0.656 $\pm$ 0.247
c10mop1_0	2.0	2.0	1.0	2.0	0.428 $\pm$ 0.392	0.468 $\pm$ 0.252	0.415 $\pm$ 0.242	0.619 $\pm$ 0.275
c10mop3_0	1.0	4.0	3.0	2.0	0.077 $\pm$ 0.245	0.520 $\pm$ 0.350	0.415 $\pm$ 0.413	0.258 $\pm$ 0.283
c10mop5_0	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.113 $\pm$ 0.197	0.266 $\pm$ 0.332	0.190 $\pm$ 0.390
c10mop8_0	1.0	3.0	2.0	2.0	0.222 $\pm$ 0.151	0.561 $\pm$ 0.217	0.541 $\pm$ 0.207	0.459 $\pm$ 0.103
citysegmop3_0	1.0	1.0	2.0	1.0	0.394 $\pm$ 0.216	0.421 $\pm$ 0.237	0.553 $\pm$ 0.285	0.382 $\pm$ 0.396
in1kmop1_0	4.0	2.0	1.0	3.0	0.764 $\pm$ 0.212	0.455 $\pm$ 0.159	0.345 $\pm$ 0.192	0.507 $\pm$ 0.117
in1kmop4_0	1.0	3.0	2.0	4.0	0.221 $\pm$ 0.178	0.466 $\pm$ 0.270	0.326 $\pm$ 0.174	0.528 $\pm$ 0.175
in1kmop7_0	1.0	3.0	2.0	2.0	0.126 $\pm$ 0.090	0.473 $\pm$ 0.248	0.455 $\pm$ 0.207	0.486 $\pm$ 0.151
Average	1.75	2.33	2.00	2.17	0.361 $\pm$ 0.140	0.468 $\pm$ 0.224	0.440 $\pm$ 0.259	0.438 $\pm$ 0.238

TABLE II: Detailed case-level results for next release problem. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
nrp-e1	1.0	2.0	3.0	1.0	0.002 $\pm$ 0.002	0.908 $\pm$ 0.040	0.962 $\pm$ 0.020	0.920 $\pm$ 0.017
nrp-e2	1.0	2.0	3.0	1.0	0.001 $\pm$ 0.001	0.931 $\pm$ 0.009	0.977 $\pm$ 0.017	0.929 $\pm$ 0.024
nrp-e3	1.0	2.0	3.0	1.0	0.000 $\pm$ 0.000	0.898 $\pm$ 0.021	0.958 $\pm$ 0.039	0.901 $\pm$ 0.033
nrp-e4	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.001	0.902 $\pm$ 0.020	0.967 $\pm$ 0.021	0.931 $\pm$ 0.025
nrp-g1	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.907 $\pm$ 0.020	0.971 $\pm$ 0.015	0.915 $\pm$ 0.029
nrp-g2	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.905 $\pm$ 0.017	0.948 $\pm$ 0.044	0.913 $\pm$ 0.025
nrp-g3	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.902 $\pm$ 0.021	0.957 $\pm$ 0.041	0.924 $\pm$ 0.034
nrp-g4	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.868 $\pm$ 0.065	0.934 $\pm$ 0.054	0.898 $\pm$ 0.041
nrp-m1	1.0	2.0	4.0	3.0	0.001 $\pm$ 0.002	0.924 $\pm$ 0.013	0.969 $\pm$ 0.013	0.929 $\pm$ 0.019
nrp-m2	1.0	2.0	3.0	1.0	0.001 $\pm$ 0.001	0.928 $\pm$ 0.021	0.974 $\pm$ 0.019	0.930 $\pm$ 0.027
nrp-m3	1.0	3.0	4.0	2.0	0.001 $\pm$ 0.001	0.955 $\pm$ 0.014	0.981 $\pm$ 0.013	0.941 $\pm$ 0.021
nrp-m4	1.0	3.0	4.0	2.0	0.001 $\pm$ 0.001	0.931 $\pm$ 0.019	0.969 $\pm$ 0.022	0.928 $\pm$ 0.016
nrp1	1.0	3.0	4.0	2.0	0.000 $\pm$ 0.000	0.894 $\pm$ 0.048	0.916 $\pm$ 0.081	0.871 $\pm$ 0.041
nrp2	1.0	4.0	3.0	2.0	0.025 $\pm$ 0.022	0.975 $\pm$ 0.017	0.902 $\pm$ 0.036	0.864 $\pm$ 0.023
nrp3	1.0	2.0	4.0	3.0	0.003 $\pm$ 0.003	0.937 $\pm$ 0.012	0.981 $\pm$ 0.011	0.947 $\pm$ 0.016
nrp4	1.0	2.0	3.0	1.0	0.000 $\pm$ 0.000	0.955 $\pm$ 0.011	0.983 $\pm$ 0.013	0.957 $\pm$ 0.009
nrp5	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.950 $\pm$ 0.012	0.982 $\pm$ 0.020	0.958 $\pm$ 0.010
Average	1.00	2.29	3.65	2.18	0.002 $\pm$ 0.002	0.922 $\pm$ 0.022	0.961 $\pm$ 0.028	0.921 $\pm$ 0.024

TABLE III: Detailed case-level results for software configuration tuning. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
dnn_adiaac	1.0	4.0	2.0	3.0	0.421 $\pm$ 0.415	0.724 $\pm$ 0.255	0.564 $\pm$ 0.277	0.594 $\pm$ 0.316
dnn_adiaac_reverse	1.0	2.0	4.0	3.0	0.003 $\pm$ 0.008	0.075 $\pm$ 0.061	0.181 $\pm$ 0.291	0.103 $\pm$ 0.044
dnn_coffee	1.0	1.0	1.0	1.0	0.314 $\pm$ 0.409	0.386 $\pm$ 0.415	0.300 $\pm$ 0.389	0.286 $\pm$ 0.369
dnn_coffee_reverse	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.135 $\pm$ 0.108	0.544 $\pm$ 0.388	0.457 $\pm$ 0.280
dnn_dsr	2.0	1.0	2.0	2.0	0.106 $\pm$ 0.205	0.162 $\pm$ 0.306	0.205 $\pm$ 0.249	0.236 $\pm$ 0.376
dnn_dsr_reverse	1.0	2.0	3.0	3.0	0.000 $\pm$ 0.000	0.089 $\pm$ 0.089	0.328 $\pm$ 0.289	0.319 $\pm$ 0.214
dnn_sa	1.0	3.0	2.0	1.0	0.270 $\pm$ 0.247	0.779 $\pm$ 0.290	0.699 $\pm$ 0.220	0.731 $\pm$ 0.230
dnn_sa_reverse	1.0	2.0	3.0	3.0	0.000 $\pm$ 0.000	0.480 $\pm$ 0.120	0.623 $\pm$ 0.285	0.623 $\pm$ 0.199
llvm	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.322 $\pm$ 0.271	0.830 $\pm$ 0.133	0.687 $\pm$ 0.230
llvm_reverse	1.0	2.0	3.0	1.0	0.150 $\pm$ 0.093	0.505 $\pm$ 0.236	0.645 $\pm$ 0.219	0.462 $\pm$ 0.286
lrzip	3.0	2.0	1.0	3.0	0.412 $\pm$ 0.142	0.453 $\pm$ 0.281	0.317 $\pm$ 0.222	0.579 $\pm$ 0.272
lrzip_reverse	1.0	4.0	2.0	3.0	0.000 $\pm$ 0.001	0.328 $\pm$ 0.306	0.037 $\pm$ 0.042	0.094 $\pm$ 0.080
mariadb	3.0	2.0	1.0	3.0	0.182 $\pm$ 0.292	0.121 $\pm$ 0.254	0.000 $\pm$ 0.000	0.430 $\pm$ 0.386
mariadb_reverse	1.0	3.0	2.0	2.0	0.039 $\pm$ 0.122	0.332 $\pm$ 0.297	0.289 $\pm$ 0.348	0.167 $\pm$ 0.219
mongodb	1.0	1.0	2.0	3.0	0.062 $\pm$ 0.080	0.071 $\pm$ 0.158	0.331 $\pm$ 0.319	0.528 $\pm$ 0.307
mongodb_reverse	2.0	1.0	3.0	4.0	0.056 $\pm$ 0.059	0.022 $\pm$ 0.047	0.449 $\pm$ 0.291	0.568 $\pm$ 0.298
storm_rs	1.0	2.0	3.0	3.0	0.058 $\pm$ 0.184	0.442 $\pm$ 0.312	0.567 $\pm$ 0.394	0.658 $\pm$ 0.355
storm_rs_reverse	1.0	1.0	2.0	3.0	0.015 $\pm$ 0.046	0.015 $\pm$ 0.046	0.259 $\pm$ 0.316	0.378 $\pm$ 0.299
storm_wc	1.0	3.0	2.0	4.0	0.000 $\pm$ 0.000	0.248 $\pm$ 0.327	0.173 $\pm$ 0.110	0.260 $\pm$ 0.163
storm_wc_reverse	1.0	2.0	4.0	3.0	0.000 $\pm$ 0.000	0.101 $\pm$ 0.136	0.392 $\pm$ 0.242	0.284 $\pm$ 0.082
trimesh	1.0	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
trimesh_reverse	1.0	3.0	2.0	2.0	0.066 $\pm$ 0.035	0.708 $\pm$ 0.175	0.669 $\pm$ 0.225	0.617 $\pm$ 0.270
vp9	1.0	2.0	3.0	3.0	0.013 $\pm$ 0.041	0.163 $\pm$ 0.036	0.198 $\pm$ 0.312	0.166 $\pm$ 0.150
vp9_reverse	1.0	2.0	1.0	1.0	0.131 $\pm$ 0.176	0.410 $\pm$ 0.364	0.326 $\pm$ 0.391	0.329 $\pm$ 0.357
x264	2.0	2.0	1.0	3.0	0.000 $\pm$ 0.000	0.887 $\pm$ 0.240	0.000 $\pm$ 0.000	0.888 $\pm$ 0.022
x264_reverse	2.0	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.590 $\pm$ 0.162
Average	1.31	2.08	2.27	2.54	0.108 $\pm$ 0.098	0.325 $\pm$ 0.197	0.362 $\pm$ 0.229	0.444 $\pm$ 0.229

TABLE IV: Detailed case-level results for software effort estimation. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
china_fold1	1.0	2.0	3.0	1.0	0.095 $\pm$ 0.073	0.640 $\pm$ 0.120	0.709 $\pm$ 0.193	0.643 $\pm$ 0.157
china_fold2	1.0	3.0	4.0	2.0	0.025 $\pm$ 0.023	0.783 $\pm$ 0.140	0.830 $\pm$ 0.100	0.721 $\pm$ 0.092
china_fold3	1.0	2.0	3.0	1.0	0.013 $\pm$ 0.019	0.362 $\pm$ 0.198	0.425 $\pm$ 0.240	0.345 $\pm$ 0.179
desharnais_fold1	1.0	3.0	2.0	1.0	0.238 $\pm$ 0.166	0.792 $\pm$ 0.151	0.651 $\pm$ 0.200	0.643 $\pm$ 0.114
desharnais_fold2	1.0	2.0	3.0	3.0	0.159 $\pm$ 0.139	0.569 $\pm$ 0.214	0.607 $\pm$ 0.166	0.622 $\pm$ 0.250
desharnais_fold3	1.0	3.0	2.0	1.0	0.107 $\pm$ 0.087	0.661 $\pm$ 0.181	0.618 $\pm$ 0.129	0.667 $\pm$ 0.169
finnish_fold1	1.0	4.0	2.0	3.0	0.033 $\pm$ 0.027	0.549 $\pm$ 0.284	0.297 $\pm$ 0.169	0.326 $\pm$ 0.109
finnish_fold2	1.0	3.0	2.0	1.0	0.099 $\pm$ 0.069	0.624 $\pm$ 0.188	0.540 $\pm$ 0.179	0.514 $\pm$ 0.129
finnish_fold3	1.0	2.0	1.0	1.0	0.027 $\pm$ 0.028	0.718 $\pm$ 0.165	0.730 $\pm$ 0.111	0.765 $\pm$ 0.183
maxwell_fold1	1.0	3.0	2.0	2.0	0.042 $\pm$ 0.024	0.873 $\pm$ 0.067	0.858 $\pm$ 0.079	0.798 $\pm$ 0.062
maxwell_fold2	1.0	3.0	2.0	2.0	0.060 $\pm$ 0.046	0.893 $\pm$ 0.066	0.894 $\pm$ 0.065	0.821 $\pm$ 0.073
maxwell_fold3	1.0	3.0	2.0	2.0	0.106 $\pm$ 0.084	0.867 $\pm$ 0.083	0.892 $\pm$ 0.071	0.790 $\pm$ 0.030
miyazaki_fold1	1.0	3.0	2.0	2.0	0.007 $\pm$ 0.005	0.637 $\pm$ 0.207	0.478 $\pm$ 0.150	0.622 $\pm$ 0.207
miyazaki_fold2	1.0	3.0	2.0	1.0	0.005 $\pm$ 0.005	0.529 $\pm$ 0.307	0.418 $\pm$ 0.126	0.573 $\pm$ 0.215
miyazaki_fold3	1.0	4.0	2.0	3.0	0.003 $\pm$ 0.002	0.601 $\pm$ 0.326	0.386 $\pm$ 0.146	0.520 $\pm$ 0.181
Average	1.00	2.87	2.27	1.73	0.068 $\pm$ 0.053	0.673 $\pm$ 0.180	0.622 $\pm$ 0.142	0.625 $\pm$ 0.143

TABLE V: Detailed case-level results for software defect prediction. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
ant-1.7_J48	1.0	3.0	2.0	4.0	0.122 $\pm$ 0.081	0.543 $\pm$ 0.121	0.456 $\pm$ 0.170	0.617 $\pm$ 0.213
ant-1.7_KNN	1.0	2.0	1.0	1.0	0.128 $\pm$ 0.062	0.463 $\pm$ 0.159	0.493 $\pm$ 0.172	0.460 $\pm$ 0.246
ant-1.7_LR	1.0	2.0	1.0	1.0	0.384 $\pm$ 0.129	0.585 $\pm$ 0.298	0.588 $\pm$ 0.272	0.569 $\pm$ 0.290
ant-1.7_NB	1.0	2.0	1.0	1.0	0.202 $\pm$ 0.110	0.511 $\pm$ 0.258	0.516 $\pm$ 0.298	0.473 $\pm$ 0.264
camel-1.6_J48	1.0	3.0	2.0	2.0	0.102 $\pm$ 0.077	0.662 $\pm$ 0.199	0.642 $\pm$ 0.217	0.578 $\pm$ 0.276
camel-1.6_KNN	1.0	2.0	3.0	1.0	0.135 $\pm$ 0.066	0.635 $\pm$ 0.179	0.685 $\pm$ 0.253	0.573 $\pm$ 0.257
camel-1.6_LR	1.0	2.0	1.0	1.0	0.178 $\pm$ 0.101	0.450 $\pm$ 0.294	0.470 $\pm$ 0.295	0.458 $\pm$ 0.334
camel-1.6_NB	1.0	2.0	1.0	1.0	0.208 $\pm$ 0.053	0.322 $\pm$ 0.164	0.279 $\pm$ 0.167	0.336 $\pm$ 0.262
ivy-2.0_J48	1.0	2.0	1.0	1.0	0.291 $\pm$ 0.148	0.750 $\pm$ 0.115	0.785 $\pm$ 0.137	0.751 $\pm$ 0.141
ivy-2.0_KNN	1.0	3.0	2.0	2.0	0.133 $\pm$ 0.084	0.539 $\pm$ 0.182	0.543 $\pm$ 0.158	0.504 $\pm$ 0.239
ivy-2.0_LR	1.0	2.0	1.0	1.0	0.113 $\pm$ 0.050	0.342 $\pm$ 0.170	0.344 $\pm$ 0.254	0.359 $\pm$ 0.195
ivy-2.0_NB	1.0	2.0	3.0	1.0	0.172 $\pm$ 0.075	0.487 $\pm$ 0.236	0.540 $\pm$ 0.292	0.549 $\pm$ 0.205
jedit-4.0_J48	1.0	2.0	3.0	3.0	0.113 $\pm$ 0.068	0.575 $\pm$ 0.235	0.683 $\pm$ 0.222	0.628 $\pm$ 0.179
jedit-4.0_KNN	1.0	2.0	1.0	1.0	0.201 $\pm$ 0.086	0.682 $\pm$ 0.183	0.645 $\pm$ 0.179	0.655 $\pm$ 0.181
jedit-4.0_LR	1.0	2.0	1.0	1.0	0.152 $\pm$ 0.097	0.613 $\pm$ 0.252	0.630 $\pm$ 0.182	0.641 $\pm$ 0.211
jedit-4.0_NB	1.0	2.0	3.0	3.0	0.135 $\pm$ 0.070	0.508 $\pm$ 0.121	0.557 $\pm$ 0.251	0.542 $\pm$ 0.183
lucene-2.4_J48	1.0	3.0	2.0	4.0	0.143 $\pm$ 0.076	0.584 $\pm$ 0.181	0.486 $\pm$ 0.171	0.593 $\pm$ 0.116
lucene-2.4_KNN	1.0	2.0	1.0	1.0	0.161 $\pm$ 0.083	0.482 $\pm$ 0.262	0.435 $\pm$ 0.219	0.463 $\pm$ 0.147
lucene-2.4_LR	1.0	2.0	1.0	1.0	0.379 $\pm$ 0.084	0.560 $\pm$ 0.258	0.527 $\pm$ 0.208	0.554 $\pm$ 0.249
lucene-2.4_NB	1.0	2.0	1.0	1.0	0.436 $\pm$ 0.091	0.624 $\pm$ 0.245	0.627 $\pm$ 0.286	0.590 $\pm$ 0.317
poi-3.0_J48	1.0	3.0	2.0	2.0	0.128 $\pm$ 0.087	0.578 $\pm$ 0.156	0.557 $\pm$ 0.192	0.516 $\pm$ 0.273
poi-3.0_KNN	1.0	2.0	3.0	3.0	0.215 $\pm$ 0.090	0.564 $\pm$ 0.238	0.575 $\pm$ 0.192	0.570 $\pm$ 0.194
poi-3.0_LR	1.0	2.0	3.0	1.0	0.264 $\pm$ 0.112	0.450 $\pm$ 0.228	0.458 $\pm$ 0.231	0.529 $\pm$ 0.159
poi-3.0_NB	1.0	2.0	3.0	1.0	0.246 $\pm$ 0.108	0.487 $\pm$ 0.213	0.513 $\pm$ 0.207	0.471 $\pm$ 0.185
synapse-1.2_J48	1.0	2.0	3.0	4.0	0.111 $\pm$ 0.079	0.471 $\pm$ 0.193	0.531 $\pm$ 0.191	0.661 $\pm$ 0.202
synapse-1.2_KNN	1.0	2.0	1.0	1.0	0.120 $\pm$ 0.093	0.583 $\pm$ 0.224	0.505 $\pm$ 0.260	0.555 $\pm$ 0.208
synapse-1.2_LR	1.0	2.0	1.0	1.0	0.217 $\pm$ 0.119	0.583 $\pm$ 0.248	0.607 $\pm$ 0.217	0.605 $\pm$ 0.206
synapse-1.2_NB	1.0	2.0	1.0	1.0	0.197 $\pm$ 0.117	0.574 $\pm$ 0.299	0.578 $\pm$ 0.228	0.599 $\pm$ 0.245
velocity-1.6_J48	1.0	3.0	2.0	1.0	0.077 $\pm$ 0.050	0.547 $\pm$ 0.168	0.485 $\pm$ 0.216	0.446 $\pm$ 0.111
velocity-1.6_KNN	1.0	3.0	2.0	1.0	0.142 $\pm$ 0.105	0.654 $\pm$ 0.081	0.506 $\pm$ 0.235	0.537 $\pm$ 0.265
velocity-1.6_LR	1.0	2.0	3.0	1.0	0.205 $\pm$ 0.075	0.347 $\pm$ 0.259	0.371 $\pm$ 0.264	0.410 $\pm$ 0.230
velocity-1.6_NB	1.0	1.0	2.0	2.0	0.308 $\pm$ 0.077	0.420 $\pm$ 0.217	0.471 $\pm$ 0.266	0.314 $\pm$ 0.186
xalan-2.6_J48	1.0	4.0	2.0	3.0	0.185 $\pm$ 0.099	0.640 $\pm$ 0.345	0.538 $\pm$ 0.240	0.613 $\pm$ 0.231
xalan-2.6_KNN	1.0	2.0	3.0	1.0	0.127 $\pm$ 0.117	0.567 $\pm$ 0.211	0.666 $\pm$ 0.251	0.499 $\pm$ 0.296
xalan-2.6_LR	1.0	2.0	1.0	1.0	0.339 $\pm$ 0.134	0.414 $\pm$ 0.300	0.419 $\pm$ 0.305	0.411 $\pm$ 0.307
xalan-2.6_NB	1.0	2.0	3.0	4.0	0.375 $\pm$ 0.107	0.447 $\pm$ 0.253	0.475 $\pm$ 0.206	0.544 $\pm$ 0.237
xerces-1.4_J48	2.0	1.0	2.0	2.0	0.253 $\pm$ 0.111	0.293 $\pm$ 0.229	0.354 $\pm$ 0.314	0.378 $\pm$ 0.288
xerces-1.4_KNN	1.0	2.0	3.0	1.0	0.138 $\pm$ 0.131	0.393 $\pm$ 0.183	0.490 $\pm$ 0.259	0.409 $\pm$ 0.176
xerces-1.4_LR	1.0	1.0	2.0	2.0	0.380 $\pm$ 0.097	0.415 $\pm$ 0.233	0.478 $\pm$ 0.272	0.456 $\pm$ 0.224
xerces-1.4_NB	1.0	2.0	3.0	1.0	0.293 $\pm$ 0.129	0.431 $\pm$ 0.292	0.526 $\pm$ 0.322	0.440 $\pm$ 0.311
Average	1.02	2.15	1.93	1.65	0.205 $\pm$ 0.093	0.519 $\pm$ 0.217	0.526 $\pm$ 0.232	0.521 $\pm$ 0.226

TABLE VI: Detailed case-level results for software product lines testing. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
7z	1.0	3.0	2.0	4.0	0.004 $\pm$ 0.002	0.440 $\pm$ 0.111	0.182 $\pm$ 0.060	0.556 $\pm$ 0.180
amazon	1.0	3.0	2.0	2.0	0.026 $\pm$ 0.013	0.792 $\pm$ 0.121	0.447 $\pm$ 0.083	0.771 $\pm$ 0.100
berkeleydbc	1.0	4.0	2.0	3.0	0.023 $\pm$ 0.020	0.769 $\pm$ 0.136	0.427 $\pm$ 0.059	0.708 $\pm$ 0.058
cocheecologico	1.0	3.0	2.0	2.0	0.036 $\pm$ 0.022	0.761 $\pm$ 0.126	0.469 $\pm$ 0.060	0.755 $\pm$ 0.099
counterstrikesimplefeaturemodel	1.0	3.0	2.0	4.0	0.053 $\pm$ 0.051	0.784 $\pm$ 0.142	0.547 $\pm$ 0.043	0.823 $\pm$ 0.111
drupal	1.0	4.0	2.0	3.0	0.049 $\pm$ 0.044	0.869 $\pm$ 0.101	0.603 $\pm$ 0.064	0.842 $\pm$ 0.062
dssample	1.0	3.0	2.0	2.0	0.015 $\pm$ 0.015	0.659 $\pm$ 0.075	0.198 $\pm$ 0.022	0.641 $\pm$ 0.197
dune	1.0	4.0	2.0	3.0	0.083 $\pm$ 0.047	0.886 $\pm$ 0.067	0.610 $\pm$ 0.054	0.779 $\pm$ 0.085
electronicdrum	1.0	3.0	2.0	4.0	0.052 $\pm$ 0.024	0.509 $\pm$ 0.069	0.315 $\pm$ 0.034	0.584 $\pm$ 0.165
hipacc	1.0	3.0	2.0	2.0	0.050 $\pm$ 0.029	0.756 $\pm$ 0.112	0.473 $\pm$ 0.072	0.773 $\pm$ 0.152
javagc	1.0	3.0	2.0	4.0	0.012 $\pm$ 0.012	0.446 $\pm$ 0.069	0.233 $\pm$ 0.051	0.568 $\pm$ 0.203
jhipster	1.0	4.0	2.0	3.0	0.058 $\pm$ 0.037	0.882 $\pm$ 0.095	0.444 $\pm$ 0.088	0.844 $\pm$ 0.110
lrzip	2.0	2.0	1.0	3.0	0.000 $\pm$ 0.000	0.615 $\pm$ 0.252	0.000 $\pm$ 0.000	0.320 $\pm$ 0.228
modeltransformation	1.0	3.0	2.0	4.0	0.086 $\pm$ 0.059	0.804 $\pm$ 0.104	0.598 $\pm$ 0.091	0.871 $\pm$ 0.067
polly	1.0	3.0	2.0	4.0	0.062 $\pm$ 0.030	0.784 $\pm$ 0.102	0.416 $\pm$ 0.060	0.827 $\pm$ 0.093
smarthomev2.2	1.0	4.0	2.0	3.0	0.058 $\pm$ 0.057	0.849 $\pm$ 0.119	0.550 $\pm$ 0.062	0.748 $\pm$ 0.045
splssimuelesnpn	1.0	4.0	2.0	3.0	0.046 $\pm$ 0.027	0.751 $\pm$ 0.177	0.461 $\pm$ 0.076	0.690 $\pm$ 0.096
videoplayer	1.0	4.0	2.0	3.0	0.065 $\pm$ 0.040	0.867 $\pm$ 0.096	0.587 $\pm$ 0.058	0.787 $\pm$ 0.075
vp9	1.0	4.0	2.0	3.0	0.068 $\pm$ 0.030	0.795 $\pm$ 0.198	0.457 $\pm$ 0.077	0.785 $\pm$ 0.053
webportal	1.0	4.0	2.0	3.0	0.041 $\pm$ 0.033	0.867 $\pm$ 0.097	0.565 $\pm$ 0.096	0.846 $\pm$ 0.084
x264	1.0	4.0	2.0	3.0	0.047 $\pm$ 0.039	0.779 $\pm$ 0.158	0.463 $\pm$ 0.069	0.700 $\pm$ 0.086
Average	1.05	3.43	1.95	3.10	0.045 $\pm$ 0.030	0.746 $\pm$ 0.120	0.431 $\pm$ 0.061	0.725 $\pm$ 0.112

TABLE VII: Detailed case-level results for software project scheduling. $\times$ indicates that the algorithm failed to obtain valid solutions on that case. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
10-10-skill-4-5	1.0	2.0	3.0	1.0	0.004 $\pm$ 0.002	0.884 $\pm$ 0.025	0.963 $\pm$ 0.022	0.893 $\pm$ 0.021
10-10-skill-6-7	1.0	2.0	4.0	3.0	0.002 $\pm$ 0.002	0.897 $\pm$ 0.024	0.977 $\pm$ 0.015	0.916 $\pm$ 0.012
10-15-skill-4-5	1.0	2.0	3.0	1.0	0.003 $\pm$ 0.003	0.929 $\pm$ 0.008	0.960 $\pm$ 0.025	0.927 $\pm$ 0.008
10-15-skill-6-7	1.0	2.0	4.0	3.0	0.001 $\pm$ 0.001	0.956 $\pm$ 0.008	0.984 $\pm$ 0.013	0.962 $\pm$ 0.007
10-5-skill-4-5	1.0	2.0	3.0	1.0	0.000 $\pm$ 0.000	0.923 $\pm$ 0.016	0.981 $\pm$ 0.016	0.919 $\pm$ 0.024
10-5-skill-6-7	1.0	2.0	4.0	3.0	0.003 $\pm$ 0.002	0.878 $\pm$ 0.039	0.951 $\pm$ 0.040	0.901 $\pm$ 0.016
20-10-skill-4-5	1.0	3.0	4.0	2.0	0.003 $\pm$ 0.003	0.940 $\pm$ 0.017	0.982 $\pm$ 0.017	0.935 $\pm$ 0.014
20-10-skill-6-7	1.0	2.0	4.0	3.0	0.002 $\pm$ 0.001	0.940 $\pm$ 0.014	0.982 $\pm$ 0.011	0.946 $\pm$ 0.015
20-15-skill-4-5	1.0	2.0	3.0	1.0	0.004 $\pm$ 0.002	0.937 $\pm$ 0.006	0.971 $\pm$ 0.020	0.940 $\pm$ 0.010
20-15-skill-6-7	1.0	3.0	4.0	2.0	0.009 $\pm$ 0.007	0.920 $\pm$ 0.009	0.963 $\pm$ 0.033	0.910 $\pm$ 0.013
20-5-skill-4-5	1.0	3.0	4.0	2.0	0.006 $\pm$ 0.004	0.873 $\pm$ 0.028	0.950 $\pm$ 0.032	0.868 $\pm$ 0.025
20-5-skill-6-7	1.0	2.0	3.0	1.0	0.003 $\pm$ 0.003	0.895 $\pm$ 0.025	0.956 $\pm$ 0.031	0.897 $\pm$ 0.026
30-10-skill-4-5	1.0	2.0	4.0	3.0	0.003 $\pm$ 0.002	0.907 $\pm$ 0.014	0.955 $\pm$ 0.032	0.915 $\pm$ 0.016
30-10-skill-6-7	1.0	2.0	4.0	3.0	0.004 $\pm$ 0.003	0.929 $\pm$ 0.013	0.975 $\pm$ 0.017	0.936 $\pm$ 0.015
30-15-skill-4-5	1.0	2.0	4.0	3.0	0.003 $\pm$ 0.004	0.896 $\pm$ 0.014	0.969 $\pm$ 0.026	0.897 $\pm$ 0.021
30-15-skill-6-7	1.0	2.0	3.0	1.0	0.006 $\pm$ 0.005	0.921 $\pm$ 0.019	0.968 $\pm$ 0.021	0.920 $\pm$ 0.017
30-5-skill-4-5	1.0	2.0	3.0	1.0	0.007 $\pm$ 0.005	0.909 $\pm$ 0.042	$\times$	0.912 $\pm$ 0.043
30-5-skill-6-7	1.0	3.0	4.0	2.0	0.004 $\pm$ 0.007	0.878 $\pm$ 0.024	0.962 $\pm$ 0.023	0.860 $\pm$ 0.016
Average	1.00	2.22	3.61	2.00	0.004 $\pm$ 0.003	0.912 $\pm$ 0.019	0.969 $\pm$ 0.022	0.914 $\pm$ 0.018

TABLE VIII: Detailed case-level results for third party library migration. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
migrationRule10_ALL	1.0	1.0	1.0	1.0	0.100 $\pm$ 0.316	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
migrationRule10_CO	1.0	1.0	1.0	1.0	0.100 $\pm$ 0.316	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
migrationRule10_DS	3.0	2.0	1.0	3.0	0.100 $\pm$ 0.316	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.200 $\pm$ 0.422
migrationRule10_MS	1.0	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule18_ALL	1.0	4.0	2.0	3.0	0.023 $\pm$ 0.073	0.618 $\pm$ 0.294	0.272 $\pm$ 0.185	0.459 $\pm$ 0.221
migrationRule18_CO	1.0	1.0	1.0	1.0	0.100 $\pm$ 0.316	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
migrationRule18_DS	1.0	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule18_MS	1.0	3.0	2.0	2.0	0.014 $\pm$ 0.045	0.414 $\pm$ 0.418	0.143 $\pm$ 0.213	0.372 $\pm$ 0.364
migrationRule1_ALL	1.0	4.0	3.0	2.0	0.079 $\pm$ 0.054	0.928 $\pm$ 0.082	0.827 $\pm$ 0.081	0.780 $\pm$ 0.125
migrationRule1_CO	1.0	4.0	3.0	2.0	0.063 $\pm$ 0.100	0.891 $\pm$ 0.109	0.858 $\pm$ 0.125	0.781 $\pm$ 0.141
migrationRule1_DS	1.0	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule1_MS	1.0	2.0	1.0	1.0	0.020 $\pm$ 0.063	0.740 $\pm$ 0.171	0.740 $\pm$ 0.117	0.730 $\pm$ 0.134
migrationRule2_ALL	1.0	3.0	2.0	2.0	0.063 $\pm$ 0.041	0.929 $\pm$ 0.052	0.939 $\pm$ 0.039	0.913 $\pm$ 0.034
migrationRule2_CO	1.0	3.0	2.0	2.0	0.139 $\pm$ 0.117	0.938 $\pm$ 0.063	0.947 $\pm$ 0.047	0.912 $\pm$ 0.055
migrationRule2_DS	1.0	4.0	3.0	2.0	0.064 $\pm$ 0.053	0.959 $\pm$ 0.040	0.926 $\pm$ 0.042	0.889 $\pm$ 0.041
migrationRule2_MS	1.0	4.0	3.0	2.0	0.089 $\pm$ 0.048	0.921 $\pm$ 0.053	0.875 $\pm$ 0.078	0.836 $\pm$ 0.100
migrationRule3_ALL	1.0	3.0	2.0	1.0	0.337 $\pm$ 0.363	0.831 $\pm$ 0.107	0.711 $\pm$ 0.322	0.707 $\pm$ 0.127
migrationRule3_CO	2.0	2.0	3.0	1.0	0.485 $\pm$ 0.337	0.590 $\pm$ 0.169	0.604 $\pm$ 0.178	0.444 $\pm$ 0.176
migrationRule3_MS	1.0	3.0	4.0	2.0	0.167 $\pm$ 0.236	0.600 $\pm$ 0.141	0.667 $\pm$ 0.157	0.500 $\pm$ 0.176
migrationRule4_ALL	1.0	3.0	2.0	1.0	0.066 $\pm$ 0.063	0.845 $\pm$ 0.130	0.763 $\pm$ 0.066	0.754 $\pm$ 0.086
migrationRule4_CO	1.0	4.0	3.0	2.0	0.000 $\pm$ 0.000	0.767 $\pm$ 0.225	0.700 $\pm$ 0.189	0.633 $\pm$ 0.105
migrationRule4_DS	1.0	3.0	2.0	2.0	0.031 $\pm$ 0.098	0.710 $\pm$ 0.177	0.743 $\pm$ 0.129	0.663 $\pm$ 0.151
migrationRule4_MS	1.0	4.0	2.0	3.0	0.000 $\pm$ 0.000	0.852 $\pm$ 0.077	0.652 $\pm$ 0.162	0.755 $\pm$ 0.130
migrationRule5_ALL	1.0	3.0	2.0	1.0	0.277 $\pm$ 0.345	0.709 $\pm$ 0.189	0.533 $\pm$ 0.168	0.506 $\pm$ 0.232
migrationRule5_CO	1.0	3.0	2.0	2.0	0.233 $\pm$ 0.225	0.900 $\pm$ 0.225	0.833 $\pm$ 0.236	0.933 $\pm$ 0.141
migrationRule5_DS	1.0	2.0	1.0	1.0	0.190 $\pm$ 0.402	0.913 $\pm$ 0.031	0.903 $\pm$ 0.000	0.903 $\pm$ 0.000
migrationRule5_MS	1.0	2.0	1.0	1.0	0.100 $\pm$ 0.316	0.500 $\pm$ 0.527	0.500 $\pm$ 0.527	0.600 $\pm$ 0.516
migrationRule7_ALL	1.0	3.0	4.0	2.0	0.102 $\pm$ 0.086	0.880 $\pm$ 0.054	0.911 $\pm$ 0.060	0.822 $\pm$ 0.072
migrationRule7_CO	1.0	2.0	4.0	3.0	0.180 $\pm$ 0.127	0.928 $\pm$ 0.046	0.971 $\pm$ 0.036	0.944 $\pm$ 0.055
migrationRule7_DS	1.0	3.0	4.0	2.0	0.132 $\pm$ 0.067	0.901 $\pm$ 0.065	0.936 $\pm$ 0.034	0.834 $\pm$ 0.064
migrationRule7_MS	1.0	3.0	4.0	2.0	0.164 $\pm$ 0.118	0.827 $\pm$ 0.085	0.905 $\pm$ 0.090	0.759 $\pm$ 0.129
migrationRule8_ALL	1.0	3.0	2.0	1.0	0.000 $\pm$ 0.000	0.820 $\pm$ 0.126	0.482 $\pm$ 0.155	0.500 $\pm$ 0.254
migrationRule8_CO	1.0	3.0	2.0	1.0	0.000 $\pm$ 0.000	0.778 $\pm$ 0.234	0.644 $\pm$ 0.250	0.667 $\pm$ 0.234
migrationRule8_DS	1.0	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule8_MS	1.0	3.0	2.0	2.0	0.000 $\pm$ 0.000	0.700 $\pm$ 0.205	0.517 $\pm$ 0.095	0.683 $\pm$ 0.183
Average	1.09	2.63	2.14	1.66	0.155 $\pm$ 0.133	0.668 $\pm$ 0.117	0.614 $\pm$ 0.108	0.614 $\pm$ 0.128

TABLE IX: Detailed case-level results for workflow scheduling. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
CyberShake_100	1.0	3.0	4.0	2.0	0.122 $\pm$ 0.084	0.612 $\pm$ 0.187	0.702 $\pm$ 0.171	0.606 $\pm$ 0.154
CyberShake_1000	1.0	2.0	3.0	1.0	0.177 $\pm$ 0.141	0.755 $\pm$ 0.083	0.824 $\pm$ 0.130	0.735 $\pm$ 0.089
CyberShake_30	1.0	3.0	4.0	2.0	0.095 $\pm$ 0.093	0.284 $\pm$ 0.276	0.343 $\pm$ 0.270	0.174 $\pm$ 0.115
CyberShake_50	1.0	2.0	3.0	1.0	0.215 $\pm$ 0.153	0.684 $\pm$ 0.113	0.754 $\pm$ 0.174	0.684 $\pm$ 0.140
Epigenomics_100	1.0	3.0	4.0	2.0	0.314 $\pm$ 0.284	0.736 $\pm$ 0.127	0.813 $\pm$ 0.098	0.667 $\pm$ 0.177
Epigenomics_24	1.0	4.0	3.0	2.0	0.300 $\pm$ 0.136	0.682 $\pm$ 0.190	0.615 $\pm$ 0.123	0.575 $\pm$ 0.208
Epigenomics_46	1.0	3.0	2.0	2.0	0.164 $\pm$ 0.176	0.844 $\pm$ 0.035	0.813 $\pm$ 0.152	0.742 $\pm$ 0.191
Epigenomics_997	1.0	2.0	1.0	1.0	0.399 $\pm$ 0.271	0.660 $\pm$ 0.065	0.637 $\pm$ 0.146	0.626 $\pm$ 0.184
Inspiral_100	1.0	2.0	1.0	1.0	0.147 $\pm$ 0.133	0.710 $\pm$ 0.082	0.740 $\pm$ 0.153	0.666 $\pm$ 0.199
Inspiral_1000	1.0	2.0	3.0	3.0	0.354 $\pm$ 0.190	0.777 $\pm$ 0.084	0.823 $\pm$ 0.103	0.816 $\pm$ 0.137
Inspiral_30	1.0	2.0	3.0	1.0	0.001 $\pm$ 0.001	0.346 $\pm$ 0.331	0.390 $\pm$ 0.319	0.474 $\pm$ 0.373
Inspiral_50	1.0	2.0	3.0	1.0	0.107 $\pm$ 0.124	0.765 $\pm$ 0.147	0.791 $\pm$ 0.129	0.754 $\pm$ 0.083
Montage_100	1.0	2.0	3.0	1.0	0.085 $\pm$ 0.061	0.626 $\pm$ 0.143	0.812 $\pm$ 0.125	0.614 $\pm$ 0.205
Montage_1000	1.0	3.0	2.0	1.0	0.169 $\pm$ 0.104	0.907 $\pm$ 0.056	0.875 $\pm$ 0.044	0.876 $\pm$ 0.066
Montage_25	1.0	2.0	3.0	4.0	0.096 $\pm$ 0.216	0.810 $\pm$ 0.046	0.829 $\pm$ 0.036	0.838 $\pm$ 0.066
Montage_50	1.0	2.0	1.0	1.0	0.095 $\pm$ 0.073	0.654 $\pm$ 0.119	0.650 $\pm$ 0.188	0.630 $\pm$ 0.167
Sipht_100	1.0	2.0	3.0	3.0	0.326 $\pm$ 0.379	0.333 $\pm$ 0.286	0.526 $\pm$ 0.390	0.446 $\pm$ 0.274
Sipht_1000	1.0	2.0	4.0	3.0	0.099 $\pm$ 0.074	0.344 $\pm$ 0.178	0.658 $\pm$ 0.258	0.479 $\pm$ 0.140
Sipht_30	1.0	3.0	4.0	2.0	0.097 $\pm$ 0.067	0.745 $\pm$ 0.127	0.758 $\pm$ 0.194	0.700 $\pm$ 0.162
Sipht_60	1.0	2.0	3.0	3.0	0.162 $\pm$ 0.108	0.461 $\pm$ 0.065	0.514 $\pm$ 0.226	0.444 $\pm$ 0.103
Average	1.00	2.40	2.85	1.85	0.176 $\pm$ 0.143	0.637 $\pm$ 0.137	0.693 $\pm$ 0.172	0.627 $\pm$ 0.162

TABLE X: Detailed case-level results for web service composition. The formate is the same as Table I.

Instance	Rank				Mean $\pm$ Std			
	MMO _b	SWAY	LINE	LITE	MMO _b	SWAY	LINE	LITE
workflow_1	1.0	3.0	2.0	1.0	0.208 $\pm$ 0.156	0.735 $\pm$ 0.152	0.723 $\pm$ 0.199	0.807 $\pm$ 0.114
workflow_10	1.0	3.0	2.0	2.0	0.432 $\pm$ 0.287	0.648 $\pm$ 0.147	0.600 $\pm$ 0.136	0.547 $\pm$ 0.188
workflow_10_reverse	3.0	3.0	1.0	2.0	0.720 $\pm$ 0.219	0.590 $\pm$ 0.158	0.526 $\pm$ 0.205	0.552 $\pm$ 0.136
workflow_1_reverse	3.0	2.0	1.0	3.0	0.642 $\pm$ 0.287	0.623 $\pm$ 0.240	0.619 $\pm$ 0.060	0.789 $\pm$ 0.151
workflow_2	1.0	3.0	2.0	4.0	0.244 $\pm$ 0.213	0.689 $\pm$ 0.128	0.618 $\pm$ 0.205	0.740 $\pm$ 0.190
workflow_2_reverse	1.0	2.0	1.0	1.0	0.586 $\pm$ 0.220	0.611 $\pm$ 0.241	0.604 $\pm$ 0.124	0.479 $\pm$ 0.266
workflow_3	1.0	2.0	1.0	1.0	0.335 $\pm$ 0.271	0.794 $\pm$ 0.121	0.783 $\pm$ 0.107	0.791 $\pm$ 0.107
workflow_3_reverse	1.0	4.0	2.0	3.0	0.490 $\pm$ 0.226	0.634 $\pm$ 0.180	0.556 $\pm$ 0.246	0.619 $\pm$ 0.229
workflow_4	1.0	3.0	2.0	2.0	0.349 $\pm$ 0.316	0.695 $\pm$ 0.138	0.558 $\pm$ 0.184	0.644 $\pm$ 0.220
workflow_4_reverse	1.0	2.0	3.0	3.0	0.553 $\pm$ 0.240	0.690 $\pm$ 0.155	0.705 $\pm$ 0.108	0.687 $\pm$ 0.186
workflow_5	1.0	3.0	2.0	1.0	0.311 $\pm$ 0.219	0.726 $\pm$ 0.191	0.703 $\pm$ 0.168	0.700 $\pm$ 0.239
workflow_5_reverse	1.0	1.0	2.0	2.0	0.498 $\pm$ 0.240	0.483 $\pm$ 0.070	0.511 $\pm$ 0.053	0.472 $\pm$ 0.185
workflow_6	1.0	3.0	2.0	1.0	0.235 $\pm$ 0.238	0.657 $\pm$ 0.145	0.618 $\pm$ 0.183	0.588 $\pm$ 0.153
workflow_6_reverse	1.0	2.0	1.0	1.0	0.572 $\pm$ 0.296	0.607 $\pm$ 0.136	0.645 $\pm$ 0.150	0.567 $\pm$ 0.136
workflow_7	1.0	3.0	2.0	2.0	0.322 $\pm$ 0.225	0.628 $\pm$ 0.237	0.543 $\pm$ 0.084	0.643 $\pm$ 0.175
workflow_7_reverse	1.0	1.0	2.0	2.0	0.712 $\pm$ 0.290	0.733 $\pm$ 0.196	0.802 $\pm$ 0.117	0.770 $\pm$ 0.165
workflow_8	1.0	2.0	3.0	1.0	0.422 $\pm$ 0.267	0.551 $\pm$ 0.196	0.580 $\pm$ 0.216	0.635 $\pm$ 0.202
workflow_8_reverse	3.0	2.0	1.0	3.0	0.584 $\pm$ 0.301	0.479 $\pm$ 0.169	0.449 $\pm$ 0.151	0.481 $\pm$ 0.145
workflow_9	1.0	4.0	2.0	3.0	0.353 $\pm$ 0.224	0.752 $\pm$ 0.184	0.582 $\pm$ 0.124	0.684 $\pm$ 0.129
workflow_9_reverse	1.0	2.0	1.0	1.0	0.593 $\pm$ 0.321	0.690 $\pm$ 0.163	0.691 $\pm$ 0.115	0.560 $\pm$ 0.134
Average	1.30	2.50	1.75	1.95	0.458 $\pm$ 0.253	0.651 $\pm$ 0.167	0.621 $\pm$ 0.147	0.638 $\pm$ 0.173